

AMAZON RETAIL ANALYTICS REPORT

Problem Statement

Amazon Fresh generates a large volume of customer, product, supplier, order, and review data every day. Managing and analyzing this data manually is difficult. Businesses need an efficient way to retrieve insights such as sales performance, customer behavior, inventory status, supplier contribution, and product ratings. SQL helps organize and analyze this data to support better business decisions.

Introduction

Amazon Fresh is an online grocery delivery service that offers a wide range of products to customers. Every purchase generates valuable data related to customers, products, suppliers, orders, and reviews.

This project uses **Structured Query Language (SQL)** to manage and analyze the Amazon Fresh database. SQL queries are used to retrieve meaningful business insights, monitor sales trends, identify top-performing products, analyze customer purchasing behavior, and improve inventory management.

Objectives

- To design and manage an Amazon Fresh database using SQL.
- To retrieve meaningful business insights from multiple tables.
- To analyze customer purchasing behavior.
- To identify top-selling and low-selling products.
- To evaluate supplier performance.
- To analyze customer reviews and ratings.
- To generate reports for business decision-making.

Dataset Description

The project consists of six tables:

- **Customers**
 - Stores customer information such as Customer ID, Name, Age, Gender, Location and Prime Membership.
 - Used to analyze customer demographics and purchasing behavior.
- **Orders**
 - Contains order-related information including Order ID, Customer ID, Order Date and Delivery Status.
 - Used to track customer orders and analyze order trends over time.
- **Order details**
 - Stores detailed information for each order, including Product ID, Quantity, and Unit Price.
 - Used to calculate sales, revenue, and product-wise order performance.
- **Products**
 - Contains product details including Product ID, Product Name, Category, Price and Stock Quantity.
 - Helps analyze product categories, inventory levels, and product performance.
- **Suppliers**
 - Stores supplier information such as Supplier ID, Supplier Name, Contact Details and Location.
 - Used to evaluate supplier contributions and supplier-wise product distribution.
- **Reviews**
 - Contains customer ratings and review comments for products.
 - Helps evaluate customer satisfaction, product quality, and overall product ratings.

Database Design

The database consists of six related tables connected through primary and foreign keys.

Relationships include:

- Customers → Orders
- Orders → Customers
- Order details → Orders, Products
- Products → Suppliers
- Suppliers → Products
- Reviews → Products, Customers

Methodology

Data Collection

- Imported CSV files into MySQL.
- Created database and tables.

Data Cleaning

- Checked missing values.
- Removed duplicate records.
- Verified primary and foreign keys.

Database Creation

- Created all tables.
- Defined constraints and established relationships.

Key Findings

- Identified top-selling products.
- Analyzed top products sold.
- Identified high-value customers.
- Determined supplier with the most products in stock.
- Compared supplier contributions.
- Ranked the customers based on their spending.

SQL Query Implementation

Executed SQL queries using:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- INNER JOIN
- Aggregate Functions (SUM, COUNT, AVG, MAX, MIN)
- CASE Statements
- Subqueries

Outcome

The SQL project successfully analyzed the Amazon Fresh database and generated meaningful business insights. It identified top-selling and low-selling products, high-value customers, supplier contributions, and inventory status. Customer purchasing behavior and product ratings were also analyzed, helping businesses understand sales performance and improve decision-making. The project demonstrates how SQL can efficiently manage large datasets and produce accurate reports for business analysis.

Conclusion

The Amazon Fresh Analytics using SQL project demonstrates the effective use of SQL for database management and business analysis. By using joins, aggregate functions, grouping, filtering, and subqueries, the project transformed raw transactional data into valuable insights. The analysis supports better decision-making in areas such as sales, inventory management, customer behavior, and supplier performance. This project highlights SQL as a powerful tool for managing relational databases and solving real-world business problems.